

VECTORS	
Term	Definition
Norm/Length	$\ v\ = \sqrt{\sum_{i=1}^n v_i^2}$
	$\ v\ = \sqrt{v^T v}$
	$\nabla \ v\ = \frac{1}{\ v\ } v^T$
Distance	$d(u, v) = \ u - v\ $
Scalar product	$u \bullet v = u^T v = \sum_k u_k v_k$

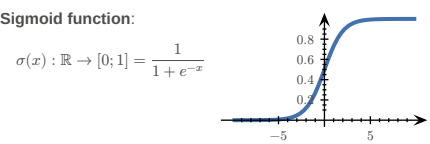
MATRICES	
Determinant:	
$\det(a) = a$	$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$
$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$	
Inverting:	
$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad - cb} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$	
Transposing:	
$\begin{pmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{pmatrix}^T = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 2 & 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 5 \end{pmatrix}^T = (1 \ 4 \ 5)$	
Matrix multiplication:	
$\begin{pmatrix} 2 & 3 & 1 \\ 4 & -1 & 7 \end{pmatrix} \cdot \begin{pmatrix} 6 & 4 \\ 1 & 0 \\ 8 & 9 \end{pmatrix} = \begin{pmatrix} 2 \cdot 6 + 3 \cdot 1 + 1 \cdot 8 & 2 \cdot 4 + 3 \cdot 0 + 1 \cdot 9 \\ 4 \cdot 6 + (-1) \cdot 1 + 7 \cdot 8 & 4 \cdot 4 + (-1) \cdot 0 + 7 \cdot 9 \end{pmatrix} = \begin{pmatrix} 23 & 17 \\ 79 & 79 \end{pmatrix}$	

INDICES	
$A \in \mathbb{R}^{n \times m}, B \in \mathbb{R}^{m \times r}, A \cdot B \in \mathbb{R}^{n \times r}$	
$(A \cdot B)_{ij} = \sum_{k=1}^m A_{ik} B_{kj} = \sum_k A_{ik} B_{kj}$	
$((A \cdot B)^T)_{ij} = (A \cdot B)_{ji} = \sum_k A_{jk} B_{ki}$	
$a^T \cdot B \cdot b = \sum_{ij} (a^T)_i B_{ij} b_j = \sum_{ij} a_i B_{ij} b_j$	
$(B \cdot a)(C \cdot a) = \sum_i (B \cdot a)_i (C \cdot a)_i$	
KRONCKER DELTA	
$\delta_{ij} = (\vec{e}_i)_j = \begin{cases} 1, & i = j \\ 0, & \text{otherwise} \end{cases}$	
$(E_{rst})_{ijk} = \delta_{ri} \delta_{sj} \delta_{tk}$	
BASIS VECTORS	
$\mathbb{R}^3$	
$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$	
UNIT MATRICES	
$\mathbb{R}^2$	
$E_{11} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, E_{12} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, E_{21} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, E_{22} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$	
AFFINE TRANSFORMATIONS	
$A_{a,M}(x) = a + M \cdot x$	
$L_A(\vec{0}) = \vec{0}$	
DERIVATIVES	
$\begin{array}{ll} 1 & \rightarrow 0 \\ x^2 & \rightarrow 2x \\ \frac{1}{x} & \rightarrow -\frac{1}{x^2} \\ e^x & \rightarrow e^x \\ \ln(x) & \rightarrow \frac{1}{x} \text{ for } x > 0 \end{array}$	$\begin{array}{ll} x & \rightarrow 1 \\ x^n & \rightarrow nx^{n-1} \\ \sqrt{x} & \rightarrow \frac{1}{2\sqrt{x}} \\ e^{-x} & \rightarrow -e^{-x} \end{array}$

$a^x \rightarrow \ln(a) \cdot a^x \text{ for } a > 0, a \neq 1$	$\ln(y \cdot x) \rightarrow \frac{1}{x} \text{ for } x > 0$
$\sin(x) \rightarrow \cos(x)$	$\log_b(x) \rightarrow \frac{1}{\ln(b) \cdot x}$
$\cos(x) \rightarrow -\sin(x)$	$\sin(2x) \rightarrow 2 \cos(2x)$
$\tan(x) \rightarrow 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$	$\cos(ax) \rightarrow -a \sin(ax)$
$\cot(x) \rightarrow -\frac{1}{\sin^2(x)}$	$\operatorname{arccot}(x) \rightarrow -\frac{1}{1+x^2}$
$\arccos(x) \rightarrow -\frac{1}{\sqrt{1-x^2}}$	$\arcsin(x) \rightarrow \frac{1}{\sqrt{1-x^2}}$
	$\arctan(x) \rightarrow \frac{1}{1+x^2}$

Term	Definition
Addition	$(f(x) + g(x))' = f'(x) + g'(x)$
Multiplication	$(\alpha \cdot f(x))' = \alpha \cdot f'(x)$
Product rule	$(f \cdot g)' = f' \cdot g + f \cdot g'$ $(f \cdot g \cdot h)' = f' \cdot g \cdot h + f \cdot g' \cdot h + f \cdot g \cdot h'$
Chain rule	$(f(g(x)))' = f'(g(x)) \cdot g'(x)$
Quotient rule	$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

FUNCTIONS OF SEVERAL VARIABLES	
Term	Definition
Real valued function	$R \subset \mathbb{R}$
Vector valued function	$R \subset \mathbb{R}^m$
A function of n variables	$D \subset \mathbb{R}^n$
Softmax function	TODO:
RSS	$RSS = \sum_i (y_i - f(x_i))^2$
n -dimensional curve	$c: \mathbb{R} \rightarrow \mathbb{R}^n$
n -dimensional hyper-surface	$s: \mathbb{R}^n \rightarrow \mathbb{R}$
Reparametrization	$c(t) \rightarrow d(s) = c(h(s))$



DERIVATIVES	
Function	Derivative
$f: \begin{cases} \mathbb{R} \rightarrow \mathbb{R}^n \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R} \rightarrow \mathbb{R}^n \\ x \mapsto y = \begin{pmatrix} \frac{d}{dx} f_1(x) \\ \frac{d}{dx} f_2(x) \\ \vdots \\ \frac{d}{dx} f_n(x) \end{pmatrix} \end{cases}$
$f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R} \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{1 \times n} \\ x \mapsto y = \nabla f = \left(\frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), \dots, \frac{\partial}{\partial x_n} f(x) \right) \end{cases}$
$f: \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^m \\ x \mapsto y \end{cases}$	$f': \begin{cases} \mathbb{R}^n \rightarrow \mathbb{R}^{m \times n} \\ x \mapsto y = J_f = \frac{\partial f}{\partial x} \Big _{x=x_0} = \begin{pmatrix} \nabla f_1(x) \\ \vdots \\ \nabla f_m(x) \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial x_1} f_1(x) & \dots & \frac{\partial}{\partial x_n} f_1(x) \\ \vdots & & \vdots \\ \frac{\partial}{\partial x_1} f_m(x) & \dots & \frac{\partial}{\partial x_n} f_m(x) \end{pmatrix} \end{cases}$

PARTIAL VS TOTAL DERIVATIVE	
TODO: example FS2024	
$f(x, y) = x^2 + 2y$	
$\frac{\partial f}{\partial x} = 2x$	
$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dx} = 2x + 2 \cdot \frac{dy}{dx} \mid (y \text{ dependent on } x)$	
GRADIENTS	
Common gradients:	
$f(x) = \ x\ ^2$	$\Rightarrow \nabla f = 2x$
$f(x) = a^T x + c$	$\Rightarrow \nabla f = a$

$f(x) = x^T A x \Rightarrow \nabla f = (A + A^T) x$	
$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow \nabla f = \frac{A + A^T}{2} x + b$	$= Ax + b \text{ if } A^T = A$
$f(x) = g(Ax + b) \Rightarrow \nabla f = A^T \nabla g(Ax + b)$	
$g: \mathbb{R}^m \rightarrow \mathbb{R}$	
JACOBIAN MATRIX	
$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2, J_f(x, y) = \begin{pmatrix} \frac{\partial x_1}{\partial x_2} & \frac{\partial y_1}{\partial y_2} \end{pmatrix}$	

Linearisation:	
$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, x_0 \in \mathbb{R}^n$	
$L_f(x) = f(x_0) + J_f(x_0) \cdot (x - x_0)$	
Chain rule:	
$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, g: \mathbb{R}^m \rightarrow \mathbb{R}^k, h: \mathbb{R}^n \rightarrow \mathbb{R}^k = g \circ f$	
$J_h = J_g(f(x)) \cdot J_f(x)$	
Common jacobians:	
$f(x) = a + M \cdot x \Rightarrow J_f = M$	
$f(x) = x \Rightarrow J_f = \mathbb{1}$	
$f(x) = \ x\ ^2 \Rightarrow J_f = (2x)^T$	

TODO: J_x for softmax $s_i = \frac{e^{w_i}}{\sum_j e^{w_j}}; J_{i,j} = s_i(\delta_{i,j} - s_j)$	
HYPER-SURFACES	
Chain rule:	
$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, g: \mathbb{R}^m \rightarrow \mathbb{R}, h: \mathbb{R}^n \rightarrow \mathbb{R} = g \circ f$	
$\nabla h(x) = \nabla g(f(x)) = \nabla g(f) \mid_{f=f(x)} \cdot \frac{\partial}{\partial x} f(x)$	
Differentiation properties:	
$f: \mathbb{R}^n \rightarrow \mathbb{R}, g: \mathbb{R}^n \rightarrow \mathbb{R}$	
$\nabla(f + g) = \nabla f + \nabla g$	
$\nabla(f - g) = \nabla f - \nabla g$	
$\nabla(f \cdot g) = g \nabla f + f \nabla g$	
$\nabla \frac{f}{g} = \frac{g \nabla f - f \nabla g}{g^2}$	

COMMUTATIVE DIAGRAMS	
$f: D \rightarrow M$ $g: N \rightarrow R$ $h: D \rightarrow R = g \circ f$	
$\begin{array}{ccc} & & g \circ f \\ & \nearrow f & \\ D & \xrightarrow{\quad} & M \xrightarrow{\quad} R \\ & \searrow g & \end{array}$	
GEOMETRY OF HYPER-SURFACES	
$f: \mathbb{R}^2 \rightarrow \mathbb{R}$	

Tangent space: $\alpha, \beta \in \mathbb{R}, T =$	
$\left\{ \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$	
Graph of a linearisation:	
$l(x) = f(x_0) + \nabla f(x_0) \cdot (x - x_0) = T_{p_0}$	
SECOND DERIVATIVE	
$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} (f(x, y)) \right) = \frac{\partial^2 f(x, y)}{\partial x^2} = \partial_{xx} f(x, y)$	
HESSIAN MATRIX	
$f: \mathbb{R}^n \rightarrow \mathbb{R}$	
$H(x, y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} = J(\nabla f)$	
$(H(x))_{ij} = \frac{\partial}{\partial x_i} \frac{\partial}{\partial x_j} f(x)$	
$\partial_i \partial_j f = \partial_j \partial_i f$	
Common Hessians:	

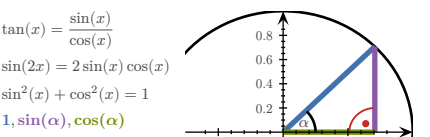
$f(x) = \frac{1}{2} x^T A x + b^T x + c \Rightarrow H_f = \frac{A + A^T}{2}$	
$f(x) = g(Ax + b) \Rightarrow H_f = A^T H_g(Ax + b) A$	$g: \mathbb{R}^m \rightarrow \mathbb{R}$
EXTREMAL POINTS	
Let $z = f(c(t))$, c a curve passing the stationary point $c_0 = (x_0, y_0)$ at $t = t_0$ and	
$\frac{d^2 z}{dt^2} = \underbrace{(\dot{c}_1(t_0), \dot{c}_2(t_0))}_{v^T} \cdot \underbrace{\begin{pmatrix} f_{xx}(c_0) & f_{xy}(c_0) \\ f_{yx}(c_0) & f_{yy}(c_0) \end{pmatrix}}_H \cdot \underbrace{\begin{pmatrix} \dot{c}_1(t_0) \\ \dot{c}_2(t_0) \end{pmatrix}}_v$	

- If $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\} (v^T H v > 0)$, then $\frac{d^2 z}{dt^2} > 0$ defines a **local minimum**
- If $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\} (v^T H v < 0)$, then $\frac{d^2 z}{dt^2} < 0$ defines a **local maximum**
- If $\forall v \in \mathbb{R}^2 \setminus \{\vec{0}\} (v^T H v < 0 \vee v^T H v > 0)$, the point $c_0 = (x_0, y_0)$ defines a **saddle point**

TODO:	
$(J_u)_{ij} = \frac{\partial u_i}{\partial x_j}, (H_g)_{ij} = \frac{\partial^2 g}{\partial x_i \partial x_j}$	
$\partial_j(u_i v_i) = (\partial_j u_i) v_i + u_i (\partial_j v_i)$	

$(v_1, v_2, v_3) \cdot \begin{pmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$	
$= (v_1, v_2, v_3) \cdot \begin{pmatrix} v_1 h_{11} + v_2 h_{12} + v_3 h_{13} \\ v_1 h_{21} + v_2 h_{22} + v_3 h_{23} \\ v_1 h_{31} + v_2 h_{32} + v_3 h_{33} \end{pmatrix}$	
$= v_1^2 h_{11} + v_1 v_2 h_{12} + v_1 v_3 h_{13} + v_1 v_2 h_{21} + v_2^2 h_{22} + v_2 v_3 h_{23} + v_1 v_3 h_{31} + v_2 v_3 h_{32} + v_3^2 h_{33}$	
$= v_1^2 h_{11} + v_2^2 h_{22} + v_3^2 h_{33} + 2v_1 v_2 h_{12} + 2v_2 v_3 h_{23} + 2v_1 v_3 h_{13}$	
Technique of completing squares	
$\underbrace{x^2}_{\sqrt{\quad}} - \underbrace{4xy}_{+2x} + 1 = (x - 2y)^2 - (-2y)^2 + 1 = (x - 2y)^2 - 4y^2 + 1$	

MISC	
$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	
TRIGONOMETRY	
x	$\begin{matrix} 0 & 30^\circ = \frac{\pi}{6} & 45^\circ = \frac{\pi}{4} & 60^\circ = \frac{\pi}{3} & 90^\circ = \frac{\pi}{2} \end{matrix}$
$\sin(x)$	$\begin{matrix} 0 & \frac{1}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{3}}{2} & 1 \end{matrix}$
$\cos(x)$	$\begin{matrix} 1 & \frac{\sqrt{3}}{2} & \frac{\sqrt{2}}{2} & \frac{1}{2} & 0 \end{matrix}$
$\tan(x)$	$\begin{matrix} 0 & \frac{1}{\sqrt{3}} & 1 & \sqrt{3} & \end{matrix}$



GRADIENT DESCENT / NEWTON'S METHOD	
$GD: x_{i+1} = x_i - \gamma \cdot \nabla f(x_i)$	
$N: x_{i+1} = x_i - \gamma \cdot (H_{f(x_i)})^{-1} \cdot \nabla f(x_i)$	
γ = step size	
termination criterion $\varepsilon > x_{i+1} - x_i $	

WORKING WITH INDICES

$x, y \in \mathbb{R}^n, \quad f: \mathbb{R}^n \rightarrow \mathbb{R}$

$$f(x) = (x - y)^T (3x - y) = \sum_{i=1}^n (x_i - y_i)(3x_i - y_i)$$

$$\partial x_i f(x) = (3x_i - y_i) + 3(x_i - y_i)$$

$$f(x) = \ln(x_1 \cdot x_2 \cdot \dots \cdot x_n) - |x|^2 = \sum_{i=1}^n (\ln(x_i) - x_i^2)$$

$$\partial x_i f(x) = \frac{1}{x_i} - 2x_i$$